20 **Ans: D** (2014 P1Q26)

- **"average" current: choose a "convenient period"**

Consider a time equal to one period where the charge that passes any given point = 1e.
Ave current = Q/t where t = T = $2\pi/\omega$, where $\omega = 4.1 \times 10^{16}$ rad s^{-1}

$= 1.60 \times 10^{-19} \div 2\pi/\omega = 1.0 \times 10^{-3}$ A